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The Difference Between Trust Measurement and Behavior: Investigating the Effect of Personalizing a Robot's Appearance on Trust in HRI

Mark Adamik[1], Karolina Dudzinska[1], Adrian J. Herskind[1], and Matthias Rehm[1]

***Abstract*— With the increased use of social robots in critical applications, like elder care and rehabilitation, it becomes necessary to investigate the user's trust in robots to prevent over- and under-utilization of the robotic systems. While several studies have shown how trust increases through personalised behaviour, there is a lack of research concerned with the influence of personalised physical appearance. This study explores the effect of personalised physical appearance on trust in human-robot-interaction (HRI). In an online game, 60 participants interacted with a robot, where half of the participants were asked to personalise the robot prior to the game. Trust was measured through a trust-related questionnaire as well as by evaluating user behaviour during the game. Results indicate that personalised physical appearance does not directly correlate to higher trust perceptions, however, there was significant evidence that players exhibit more trusting behaviours in a game against a personalised robot.**

I. INTRODUCTION

The ability to develop and maintain trust in an interpersonal relation is critical when interacting with other people. Trust can encourage cooperation, reduce the number and severity of harmful conflicts, and increase the effectiveness of crisis response [13]. With robots moving out of restricted factory settings, laypersons are also expected to interact with them, but whether robots will be accepted will strongly depend on whether people trust them or not. So far, research on trust in interactions with technology has focused on goal achievement and task efficiency, but with laypersons – or rather interaction partners – additional factors of trust have to be taken into account. Especially trust-building and maintenance (and possibly recovery) over time are so far unexplored.

Previous research relies largely on post-experimental questionnaires to assess the beliefs and expectations involved in trust (e.g. [18], [29]). While this is a good starting point for assessing the user's trust level towards the robot post-interaction, it does not indicate the precise characteristics of the robot that result in these trust levels. It also does not take into account that there might be a difference between self-reported trust levels and the actual behaviour towards the system.

In principle, there are two possibilities for influencing a user's trust in a robotics system [20]: i.) static characteristics like the robot design, ii.) dynamic characteristics such as the robot's behaviour. In this paper, we are focusing on the first aspect and investigate how the personalization of the appearance of a robot influences the user's perceived trust level and the user's behaviour during the interaction with the robot.

[1]The authors are with the HRI-lab at the Technical Faculty of IT and Design, Aalborg University, 9000 Aalborg, Denmark matthias@create.aau.dk

II. RELATED WORK

Trust is a psychological concept with many dimensions [22]. A trust relation involves someone that trusts (the trustor, in our case the user) and someone that is trusted (the trustee, in our case the robot). The general tendency to trust someone else is called propensity to trust [17]. The degree to which one has developed a propensity to trust is based on previous experiences and general characteristics of the trustor, e.g. the appearance of the trustee lets the trustor evaluate the trustworthiness of their counterpart. Trustworthiness is thereby, as opposed to propensity to trust, not target-agnostic, but rather a subjective evaluation of the degree to which the counterpart is worthy of trust, and can be subject to change as the interaction progresses [11]. Lastly, trust behaviours are the actions that the trustor engages in and do not necessarily reflect the perceived subjective level of trustworthiness.

Trust is also not static, but subject to change as a relationship evolves. On initial encounter, prior to any interaction, dispositional trust is based on the trustor's propensity to trust and the initial trustworthiness of the counterpart, but when an interaction progresses and relationships evolve, it shifts toward historical trust, which is no longer exclusively derived from prior knowledge but increasingly from the interaction itself [24]. Trust always involves the willingness of the trustor to make themselves vulnerable to the trustee's actions, based on the beliefs about that person's trustworthiness. In turn, a trustee acts trustworthy if they voluntarily decide not to take advantage of the trustor's vulnerability [6]. Additionally, in longer trust-based interactions the roles of trustor and trustee are often rotating, which gives it a reciprocal nature where a trustee not only has to consider the direct effect of their behaviour but also the repercussions it could evoke.

In automation, trust can be defined as the operator's perception of the machine's competence [25] and the assumption that the system adequately and safely performs its tasks. As Muir and Moray show, there is a high positive correlation between an operator's trust in the machine and their willingness to use it. Trust in automation is thus based on the expectation that the system will do the appropriate action for the current step in the work process [10], leaving aside the more social aspect of e.g. moral social order or expectations regarding the social environment [4].

While human-automation-interaction is thus still unilateral and requires relatively little vulnerability of the trustor,

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human-robot-interaction (HRI) increasingly resembles types of human-human-interaction. One class of robots, which is used in close contact with humans and where the user often displays vulnerability, is social robots [16]. Social robots are used in healthcare [19], education [9], or the hospitality sector [1]. Given the importance of the tasks where social robots are applied, combined with the finding that trust is closely correlated to the users' willingness to use the robot [25], it is important to research the factors influencing trust in HRI.

Merrit and Ilgen [24] find that individuals with a higher propensity to trust exhibit higher dispositional trust to the machine and in case of well-functioning increase their trust. On the other hand, they are also in danger of reacting more negatively to a malfunctioning of the machine than individuals with a lower propensity to trust. Following this, Chen et al.[7] considers the potentially greater loss of trust and successfully applies the strategy that a supervised robot, which is initially highly trusted, performs difficult tasks first, while a robot, which gets low initial trust, first does easy tasks to build trust. Dispositional trust is also influenced by the expectations of the user regarding the robot's capabilities. Since users form certain expectations of robots based on their physical appearance, it is possible that a humanoid form can set the expectations of the user too high, which could lead to a loss of trust if the robot cannot perform accordingly. Therefore, it is necessary to balance the potential advantages of humanoid appearance with the resulting expectations towards its performance [14]. Furthermore, Schaefer [29] finds that preconceived ideas about the robot's intelligence levels are based on the appearance and formed before any interaction takes place, and that only a limited number of studies so far have focused on the impact of the robot design.

While dispositional trust is mainly based on experience and the robot's appearance, history-based trust largely depends on the behaviour of the robot during the interaction. In an ongoing interaction, the actors can either make their decisions outcome-based or intention-based, where the former means that only the directly calculable outcome is considered while the latter means that the intentions of the other actors are considered as well to assume their reactions and the consequences of these. McCabe [23] found that human interaction largely consists of intention-based acting. These findings show that humans are prone to attribute each other's motivations based on behaviour and it has to be considered whether a humanoid robot can invoke similar assumptions. As Baker et al. [3] find, humans do perceive motives and characteristics when interacting with robots based on the robot's movements and actions, which can lead to humans attributing the robot intentions or to wrongly assume the intended purpose of the robot. This can have a great effect on the perceived level of trust and is exclusively based on the image the human has of the robot.

To be able to make predictions about trust changes over time, the factors influencing trust between humans and robots must be determined. Hancock et al. [15] developed a model where the components of trust have been classified into three different categories, human-related, robot-related, and environment-related. The authors found that robot performance-related factors were the greatest contributors of trust. Schaefer [29] created an updated version of the model that includes antecedents of trust as well as other influences such as gender and age.

Considering that both the appearance and the behaviour of a robot have a direct influence on the trust experienced by the user [15], it is relevant to look at the research, which has tried to adapt the appearance and behaviour to the preferences of the user to build and sustain trust during an interaction. Though not directly referring to trust, Churamani et al. [8] find that the personalized interaction capabilities of a NICO robot greatly improved how likeable and intelligent it was perceived in personalized conversations. Still, they also find that not all attributes were positively correlated since the perceived safety of the robot decreased. Langer et al. [21] investigate the effect of personalized behaviour in robots used in rehabilitation and find that to design trustworthy robots, it is necessary to achieve a high level of personalization, expressive features, and good strategies for handling errors.

To sum up, while there is a body of research concerned with the personalization of robot behaviour in relation to the perceived trust of the user, there also seems to be a lack of studies looking into personalized physical appearance and its influence on trust. Taking a basis in the research of [29] showing the influence of design and appearance on trust as well as [21] and [8] finding that personalized behaviour increases trust, this paper investigates what influence personalized physical appearance has on trust.

III. Investigating trust in HRI through Robot Personalization

Personalizing the physical appearance of the robot is expected to have a direct positive effect on the perceived trustworthiness of the robot and, thereby, the dispositional trust that the user is experiencing prior to any interaction. Hence an increase in trust should be observable both through a post hoc questionnaire and more trusting behaviour during the interaction. To create a scenario where the participants could interact with the robot in a trust-related way, we decided to use a game scenario, which requires both players to assess whether they can trust the opponent and at the same time to make the decision how much they want to risk their own trustworthiness. To this end, a webpage was designed allowing to personalize a Nao robot and subsequently play a game of dice with the robot.

A. Personalisation

The participants in the experimental group could customize three different colours for the robot, the primary colour of the robot, the colour of the joint protectors, as well as the eye colour. There are 12 different colours for the primary and secondary colours and 6 different colours for the eyes. Furthermore, the user got to change the name of the robot. The personalization page is shown in Figure 1. Members of the control group were not able to personalize

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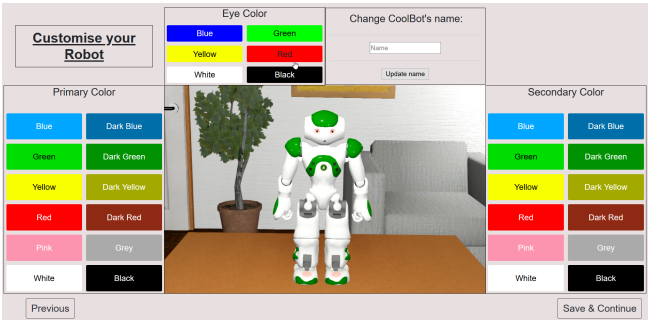

Fig. 1. Personalization web page for the robot. The user can change three different colours and the name of the robot.

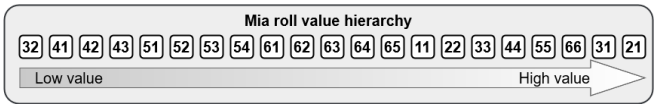

Fig. 2. Ranking order of scores

the robot, instead, they were shown a short introductory video of the robot together with a neutral descriptive text introducing the *Nao* robot. This control condition was created to ensure equal exposition to the robot system that is used in the game prior to the first trust measurement.

B. Game description

Participants played a game called Mia. The game consists of a series of rounds, where in each round players take turns rolling two dice and secretly checking the score. A player can choose to announce the true score or make up a result. The announced value however has to be higher or equal to the value previously declared by the opponent. The ranking order of the rolls is shown in Figure 2. The next player can either believe the announced result and let the game continue, or distrust and check the opponent, in which case the dice are revealed and the round ends. Depending on whether the announced value was true or not the winner of the round is declared.

The following is a short example of a game move: **Player A** rolls 51, but decides to lie and declares 22. **Player B** trusts player A, rolls 55 and declares 55. **Player A** decides to check player B, the rolled value (55) is revealed. Because player B told the truth, he wins this round.

In case one of the players declares a value of 21, the opponent cannot roll the dice in their turn. He can either trust the player, the dice will not be revealed and the player gets a point and wins the round, or he can check the player. If the declaration turns out to be true, the player gets two points. If the declaration was a lie, the opponent gets two points and wins the round. When the game ends the player with the highest number of points is declared the winner.

Mia was selected for this experiment for several reasons. Firstly, the game is quite popular and can be quickly understood by novices. Secondly, when the robot lies or tells the truth, the participant can reasonably deduce that the robot made that decision based on its ”intentions” to win the game, not because of a malfunction or any mechanical errors. Because of that, the behaviour of the robot can influence the trust of the participant. Lastly, since the players take turns and both have to trust and display trustworthiness, the game is of reciprocal nature where the long term effects of a decision have to be taken into consideration as well.

IV. Trust Experiment

A. Experimental Design

A between-group design was chosen for this study in order to manipulate dispositional trust through personalisation and compare the resulting behaviours and trust scores to the control group. The experimental group gets to personalise the robot prior to the interaction, while the control group plays against a standard version of the robot. The robots do not differ in their actual behaviour during the interaction. The historic trust of the user towards the robot is thus not dependent on the robot behaviour but only on the user. To control for user and robot behaviour, the game was fixed, i.e. participants experienced the same game situations and results of their dice rolls. The only variation thus comes from personalizing the robot or not.

B. Hypotheses

The goal of this experiment is to find whether the personalised physical appearance of a robot leads to increased trust in the robot similar to the way personalised behaviour was shown to do. One way to measure trust is through questionnaires determining the perceived (subjective) trust towards the robot as a ”trust score”.

> *H1: Interacting with a personalised robot will be positively related to higher trust scores.*

Another way to investigate trust is to look at the behaviour of the player during the game. A well-established way to measure trust in human interaction is through economic-decision games or trust games (e.g. [5]). The major components of trust games are trust actions, or trust behaviour, exhibited by the players. While many trust surveys are already well established, it can be challenging to define how exactly trust reflects in behaviour. Trust actions can be defined as risk-taking behaviours indicating a trusting relationship [26], which is also often related to the need of making oneself vulnerable to the trustee [12]. In the game of Mia, the aspect most closely related to trusting and making oneself vulnerable is whether to trust what the robot announced or to check it, where a trusting relationship would be indicated by a reluctance to mistrust the robot. Thus, it is expected that participants interacting with a personalised robot are more reluctant to mistrust the robot and, therefore, game rounds would on average end in higher rolled values.

> *H2: Interacting with a personalised robot will be positively related to trusting the robot in high stakes situations.*

To not only look at trust behaviour from a direct perspective of trust or mistrust, the experiment also considers reciprocity as a concept involved in trust relationships. In

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trust games, reciprocity was found to be an important factor [23] and it has been shown that humans behaved equally reciprocal towards robots and to other humans [27]. In the game of Mia, one action that could be interpreted in the context of reciprocity, is how often the user is lying to the robot even though they do not have to. If the user is lying without need, they run the risk of being exposed, which might lead to the robot acting reciprocally. Therefore, it can be assumed that a user who feels a stronger trust relationship to the robot, is less likely to risk this trust and abuse the robot's vulnerability by lying. Thus, it can be expected that the participants interacting with a personalised robot are more truthful than the ones interacting with an unpersonalised robot.

H3: Interacting with a personalised robot will be positively related to telling the truth when possible.

C. Measurements

The dependent variables are measured with a mix of subjective questionnaires and objective game data. Questionnaires provide the following measurements:

- **Initial and final trust:** Trust scores were calculated for all participants both before and after the interaction using the Trust Perception Scale - HRI [28].
- **Intermediate trust:** To measure the effect of the act of personalization itself, three items of the Trust Perception Scale HRI were administered.

In the context of this experiment, we define trusting behaviour as the reluctance of the player to distrust the robot's dice declarations and their honesty towards the robots. The following game variables were measured to quantify this trusting or distrusting behaviour of the participants:

- **Median trust limit:** For each round it is checked after which roll declaration the user decides to distrust the robot and check it. For each participant, the median of those scores is calculated. Before calculating the median, the values declared by the user and the robot are transformed to ranks, e.g. the value 55 is in the 17 th place in the ranking order of all possible rolls, so it will be coded as 17 in a range from 1-21.
- **All-in decision:** In the 4th round of the game the robot starts by declaring 21. The participant cannot roll the dice in this turn, they can only trust the robot, or check if it lied. The percentage of players in the respective groups who decided to trust the robot was calculated.
- **Number of unnecessary lies:** For each participant, it was checked how often they told the truth when they were not in need of lying. The result is given as a percentage.

D. Participants

Due to technical errors, some of the data from questionnaires was not saved, and, therefore, we decided to exclude these data sets from our analysis. In the end, a total of 60 participants (29 male, 31 female) with a mean age of 28.22 years (SD = 7.71), spanning from ages 19 to 55,

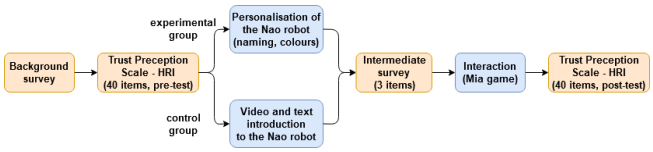

Fig. 3. Flowchart of the experiment

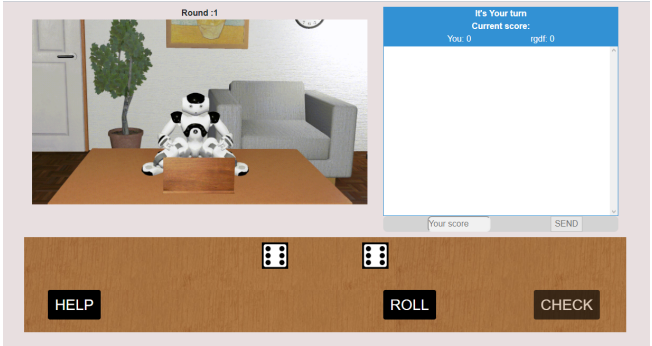

Fig. 4. Setup of the Mia game in the experiment.

participated in the experiment with complete data sets. The participants were of 23 different nationalities, with the largest group coming from Denmark (26.6%). 63.3% previously interacted with a physical robot, while only 50% have done so in a simulation. Participants were found through social media channels, like Facebook groups and did not receive any monetary compensation.

E. Apparatus

Trust scores are measured in the pre- and post-test using the Trust Perception Scale - HRI [28]. It measures the participant's response to 40 trust-related questions on an 11-point scale representing a $0 - 100\%$ range for the frequency with which the robot will display a specific behaviour. The questionnaire provides a trust score, which is calculated as a mean of the participant's response to the 40 questions. The intermediate survey consists of three selected items from the Trust Perception Scale - HRI, which were deemed as especially relevant in measuring trust in the context of a trust game. The three items are "Act Consistently", "Tell Truth", and "Behave Predictably". The intermediate survey is given to the participants in both groups right after the personalisation and general robot introduction, respectively, to measure any possible direct effect of the personalisation prior to the interaction. A free, open-source simulation software, *WeBots* was used to simulate the *Nao* robot and its environment, which resembled a living room. The robot behaviour was programmed using *Python*. The robot behaviour was identical in both groups and follows a predefined script.

F. Experimental procedure

Figure 3 presents an overview of the experimental procedure. Participation took place online and all data was collected via a web page and was stored in anonymised form. When entering the page, each participant was randomly assigned to the experimental group or the control group. The

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participants did not know which group they were assigned to or that there even were different groups. Furthermore, the participants were not informed that the experiment was investigating trust but only got to know that they would interact with a robot. Following an introduction page, demographic background data was collected from all participants which included age, gender, nationality, and level of education. The participants were also asked about their previous experiences with robots, for example, whether they interacted with a real or a simulated robot before. Subsequently, all participants were introduced to the robot they were going to interact with. Next, the participants were asked to complete the Trust Perception Scale - HRI. After completing the scale, members of the experimental group got to personalise the robot before interacting with it. Members of the control group were shown a short introductory video of the robot together with a neutral descriptive text introducing the *Nao* robot. Before playing the game, participants were asked to fill out a short intermediate survey. After completing the intermediate survey, the rules of the game were introduced and the participants were able to play Mia with the robot. In Figure 4 the game window is presented. The participant is not aware of the number of rounds they are about to play. At the beginning of the game, they were informed that each round has a 20 % probability of being the last one. However, the number of games each participant plays is set to seven. Finally, after playing the game with the robot, the participants filled out a second Trust Perception Scale - HRI, after which they could provide feedback. The experiment took approximately 15 minutes to complete.

G. Results

The questionnaire data is treated as non-parametric and analyzed using a Wilcoxon test. Results from the pre-/post-test trust scores for the experimental group ($Mdn = 0.75$) were not significantly larger than for the control group ($Mdn = 0$), $W = 433, p = 0.404$, one-tailed. The effect was small-sized $\rho = -0.155$. The same can be observed when only considering the 14 special trust related items, where the differences between pre-test and post-test in the experimental group ($Mdn = 1.07$) and control group ($Mdn <= 0.001$) were not significant, $W = 448.5, p = 0.4941$, one-tailed. The effect was small-sized $\rho = -0.187$. To capture whether the personalisation of the robot's appearance had a direct influence on the perceived trust, the three-item intermediate survey was compared to the pre-test results. There was no significant deviation in the differences in trust scores between the experimental group ($Mdn = 4.5$) and the control group ($Mdn = 2$), $W = 419.5, p = 0.3287$, one-tailed. The effect was small-sized $\rho = -0.379$.

Additionally to the subjective measurements of perceived trustworthiness through the survey, an objective, behaviour-based measurement of trust was collected, by analysing the decisions made in the game. The median trust limit for the experimental group ($Mdn = 18$) was significantly larger than for the control group ($Mdn = 15.42$), $W = 300.5, p = 0.0137$. The effect was small-sized $\rho = -0.193$. The data for all-in-decision shows that in the experimental group 40 % of users trusted the robot when it declared 21, and in the control group, 37 % did. No statistical tests were run on this data. In relation to the number of unnecessary lies, the percentage of users telling the truth in the experimental group was not significantly higher than in the control group $X2(11, N = 60) = 5.92, p = 0.879$ $\rho = 0.314$.

H. Discussion

No significant differences were found between the experimental group and the control group for the trust questionnaire. Data from this experimental setup did not provide sufficient evidence to find a positive relation between the possibility to personalise the physical appearance of the robot for the interaction and higher trust scores. Therefore, hypothesis 1 is not supported. For extracting a variety of objective data from the game, the concept of trusting behaviour has been looked at from two different perspectives, as mentioned in Section IV-B: 1) Direct trust/mistrust in the robot's actions, measured as median trust limit, and 2) General reciprocity influencing behaviour in a trust-based interaction, shown in how often the user lies to the robot. When looking at the results from the data analysis, it can be seen that the behaviour of the participants in relation to directly trusting the robot, was significantly different in the experimental group than in the control group, as users in the former group trusted the robot until a higher declared number. Therefore, hypothesis 2 is supported. The other element of trust behaviour, how often the user lies to the robot, has not shown any significant differences in the data between the experimental and control group. Therefore, hypothesis 3 is not supported.

To sum up, while we could not show an effect of personalizing the physical appearance robot on the subjectively perceived trust level, participants nevertheless exhibited increased trusting behaviour toward the personalized robot during the interaction. This shows that post-hoc questionnaires of trust in a system are not necessarily good predictors of user behaviour and should be treated with caution. Trust is an important factor in HRI, especially when robots start to increasingly interact with laypersons and not educated robot operators. One goal is to adapt robot behaviour to match or adjust a user's trust level over time. Until now, measurements are often based on either post hoc questionnaires or task success criteria. While we equalise pre-interaction exposure time for both conditions, we cannot with confidence say whether the difference in type of exposure can be adequately compared or if either of them had a bigger impact on the trust behaviour of the participant. The fact that users were interacting with a simulation instead of a real physical robot might also influence their behaviour. All questionnaires used in this experiment relate to HRI, and it's uncertain to what extent it applies to an interaction with a virtual agent. Differences between the perception of interactions in the context of trust between a robot and a virtual agent could be further investigated in future studies.

Given the novelty of this study to investigate the influence

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of personalised physical appearance on trust perception and behaviours, it is too early to draw final conclusions from this experiment. Since trust is a multidimensional concept, customizing the robot´s appearance could affect many different aspects and as Ajzen [2] points out many of them are separate constructs, like trust *intentions* and trust *behaviour*. Therefore, it would also not be obvious to conclude that a customized robot, which users claim to trust, actually gets treated with more trusting behaviour. While several studies found that personalised behaviour can increase the trust in a robot, we have shown here that personalized physical appearance has an effect on the trusting behaviour of the user. Higher trust in a robot has been proven to both make users more willing to use the robot and to benefit the users more during future interaction; and while this study did not show a significant increase in perceived trust due to personalised physical appearance, there was a significant difference in how much the user trusted the robot during the interaction, which points towards a similar effect.

V. Conclusion

In this paper, we presented a study investigating the effect of personalizing a robot's physical appearance on trust. Trust was measured in two ways, by a standard HRI trust questionnaire and by interaction measurements derived from game logs. Results show that personalization does not affect subjectively perceived trust levels, but significantly affect the behaviour of the users. Users playing with a personalized robot exhibit more trusting behaviours during the game. Thus, we can conclude that there is an effect of the personalization process, similar to what has been shown for personalized robot behaviours. But the results also show a discrepancy between the subjective and objective measurements and suggest further investigations in observable trust measurement for robust predictions of user trust levels

References

[1] Savioke announces first las vegas installation and new relay features. https://www.savioke.com/blog/2018/1/8/gpwkugbcn9xev881b1wpko1p63ekai. Accessed: 2020-09-17.

[2] Icek Ajzen. The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2):179 – 211, 1991. Theories of Cognitive Self-Regulation.

[3] Anthony L. Baker, Elizabeth K. Phillips, Daniel Ullman, and Joseph R. Keebler. Toward an understanding of trust repair in human-robot interaction. *ACM Transactions on Interactive Intelligent Systems*, 8(4):1–30, November 2018.

[4] Bernhard Barber. *The Logic and Limits of Trust*. Rutgers University Press, New Brunswick, NJ, 1983.

[5] Joyce Berg, John Dickhaut, and Kevin McCabe. Trust, reciprocity, and social history. *Games and Economic Behavior*, 10(1):122–142, July 1995.

[6] Iris Bohnet. *trust in experiments*, pages 253–257. Palgrave Macmillan UK, London, 2010.

[7] Min Chen, Stefanos Nikolaidis, Harold Soh, David Hsu, and Siddhartha S. Srinivasa. Planning with trust for human-robot collaboration. *CoRR*, abs/1801.04099, 2018.

[8] Nikhil Churamani, Paul Anton, Marc Brügger, Erik Fließwasser, Thomas Hummel, Julius Mayer, Waleed Mustafa, Hwei Geok Ng, Thi Linh Chi Nguyen, Quan Nguyen, Marcus Soll, Sebastian Springenberg, Sascha Griffiths, Stefan Heinrich, Nicolás Navarro-Guerrero, Erik Strahl, Johannes Twiefel, Cornelius Weber, and Stefan Wermter. The impact of personalisation on human-robot interaction in learning scenarios. In *Proceedings of the 5th International Conference on Human Agent Interaction*. ACM, October 2017.

[9] City of Tampere. Talking robots to be piloted in preschools and first years of primary education. https://www.tampere.fi/en/city-of-tampere/info/current-issues/2018/03/19032018_1.html. Accessed: 2020-03-31.

[10] Mark Daly David P. Biros and Gregg Gunsch. The influence of task load and automation trust on deception detection. *Group Decision and Negotiation*, 13(2):173–189, 2004.

[11] S. Kate Devitt. Trustworthiness of autonomous systems. In *Foundations of Trusted Autonomy*, pages 161–184. Springer International Publishing, 2018.

[12] Anthony M. Evans and William Revelle. Survey and behavioral measurements of interpersonal trust. *Journal of Research in Personality*, 42(6):1585 – 1593, 2008.

[13] D. Gambetta. Can We Trust Trust? *Trust: Making and Breaking Cooperative Relations, electronic edition, Department of Sociology, University of Oxford*, page 213–237, 2000.

[14] Randy Gomez. Meet haru, the unassuming big-eyed robot helping researchers study social robotics. *IEEE Spectrum*, 2020.

[15] Peter A. Hancock, Deborah R. Billings, Kristin E. Schaefer, Jessie Y. C. Chen, Ewart J. de Visser, and Raja Parasuraman. A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, 53(5):517–527, 2011.

[16] F. Hegel, C. Muhl, B. Wrede, M. Hielscher-Fastabend, and G. Sagerer. Understanding social robots. In *2009 Second International Conferences on Advances in Computer-Human Interactions*, pages 169–174, 2009.

[17] Sarah A. Jessup, Tamera R. Schneider, Gene M. Alarcon, Tyler J. Ryan, and August Capiola. The measurement of the propensity to trust automation. In *Virtual, Augmented and Mixed Reality. Applications and Case Studies*, pages 476–489. Springer International Publishing, 2019.

[18] Jiun-Yin Jian, Ann M. Bisantz, and Colin G. Drury. Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, 4(1):53–71, 2000.

[19] Reza Kachouie, Sima Sedighadeli, Rajiv Khosla, and Mei-Tai Chu. Socially assistive robots in elderly care: A mixed-method systematic literature review. *International Journal of Human-Computer Interaction*, 30(5):369 – 393, 2014.

[20] Bing Cai Kok and Harold Soh. Trust in robots: Challenges and opportunities. *Current Robotics Reports*, 1(4):297–309, September 2020.

[21] Allison Langer, Ronit Feingold-Polak, Oliver Mueller, Philipp Kellmeyer, and Shelly Levy-Tzedek. Trust in socially assistive robots: Considerations for use in rehabilitation. *Neuroscience & Biobehavioral Reviews*, 104:231–239, September 2019.

[22] Roger C. Mayer, James H. Davis, and F. David Schoorman. An integrative model of organizational trust. *The Academy of Management Review*, 20(3):709, July 1995.

[23] Kevin A. McCabe, Mary L. Rigdon, and Vernon L. Smith. Positive reciprocity and intentions in trust games. *Journal of Economic Behavior & Organization*, 52(2):267–275, October 2003.

[24] Stephanie M. Merritt and Daniel R. Ilgen. Not all trust is created equal: Dispositional and history-based trust in human-automation interactions. *Human Factors*, 50(2):194–210, 2008. PMID: 18516832.

[25] Bonnie M. Muir and Neville Moray. Trust in automation. part ii. experimental studies of trust and human intervention in a process control simulation. *Ergonomics*, 39(3):429–460, 1996. PMID: 8849495.

[26] Madan Pillutla, Deepak Malhotra, and J. Murnighan. Attribution of trust and the calculus of reciprocity. *Journal of Experimental Social Psychology*, 39:448–455, 09 2003.

[27] Eduardo Benítez Sandoval, Jürgen Brandstetter, Mohammad Obaid, and Christoph Bartneck. Reciprocity in human-robot interaction: A quantitative approach through the prisoner's dilemma and the ultimatum game. *International Journal of Social Robotics*, 8(2):303–317, December 2015.

[28] Kristin E. Schaefer. The perception and measurement of human-robot trust. 2013.

[29] Kristin E. Schaefer. Measuring trust in human robot interactions: Development of the "trust perception scale-HRI". In *Robust Intelligence and Trust in Autonomous Systems*, pages 191–218. Springer US, 2016.

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